Time Series Analysis of the EUR/DZD Black Market Exchange Rate

TSAC Individual Project 2025/2026

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Abstract

This project analyses the monthly EUR/DZD black-market (parallel) exchange rate from January 2016 to March 2026 ($n = 123$ observations). Daily sell-rate data were aggregated to monthly averages to obtain a cleaner series for classical time-series modelling. Exploratory analysis revealed a strong upward trend, visible annual seasonality, and several sharp shocks linked to political events, COVID-19, policy changes, and fiscal pressures. Box–Cox analysis indicated that no variance-stabilising transformation was needed, while the Augmented Dickey–Fuller test showed that the original series was non-stationary and that first differencing ($d = 1$) was sufficient to achieve stationarity. ACF/PACF inspection and BIC-based `armasubsets()` selection identified ARIMA(1,1,0) as the best parsimonious non-seasonal model. However, diagnosis showed that this ARIMA specification is not good, and forcasting converges too quickly to a flat path failing to reproduce the yearly pattern in the data. We therefore extended the model to SARIMA(1,1,0)(1,1,0)$_{12}$, which captures the seasonal structure and produces materially more realistic forecasts. Residual diagnostics for the SARIMA model indicate no statistically significant remaining autocorrelation at lag 12 (Ljung–Box $p = 0.099$), but the Shapiro–Wilk test rejects normality ($p = 2.19 \times 10^{-7}$), reflecting the heavy-tailed shocks present in this informal market. A 10-step hold-out comparison showed that the SARIMA model substantially improves forecast accuracy relative to the pure ARIMA benchmark. Overall, SARIMA(1,1,0)(1,1,0)$_{12}$ provides the most credible forecasting tool in this analysis, although its prediction intervals should be interpreted cautiously because exogenous shocks remain the main source of error.

Contents

1 Introduction **1**

2 Model Specification **1**

2.1 Data Loading and Preparation 1

2.2 Annotated Time Series with Key Events 1

2.3 Variance Stabilisation — Box–Cox Transformation 3

2.4 Stationarity Testing — ADF Test 4

2.5 Differencing 4

2.6 ACF and PACF Analysis 5

2.7 Automated Model Selection — `armasubsets` 6

2.8 Baseline Non-Seasonal Model 6

3 Fitting and Diagnostics **6**

3.1 Baseline ARIMA(1,1,0) Fit 6

3.2 Seasonal Extension to SARIMA(1,1,0)(1,1,0)$_{12}$ 7

3.3 Residual Diagnostics 7

3.4 Formal Diagnostic Tests 8

4 Forecasting **8**

4.1 Forecast Results 8

4.2 Forecast Plots 9

5 Discussion **9**

5.1 Main Conclusions 9

5.2 Problems Encountered 10

1 Introduction

Algeria operates a dual exchange-rate system. The official EUR/DZD rate is set by the Bank of Algeria, while a significantly higher "parallel" (black-market) rate reflects the true willingness of citizens and businesses to pay for foreign currency. The gap between the two rates has widened dramatically over the past decade, driven by oil-price shocks, political crises, pandemic restrictions, and government policy experiments. Understanding the dynamics of this parallel rate is important for several reasons: it affects the purchasing power of ordinary Algerians, and it directly influences import costs for goods ranging from vehicles to electronics.

The dataset/API used in this project consists of daily EUR/DZD sell rates quoted on the Algerian black market, spanning from January 2016 to March 2026. The raw data contain over 3 300 daily observations, many of which are identical on consecutive days (creating a "staircase" pattern). To obtain a cleaner signal and a sample size amenable to time-series modelling, we aggregate the daily values into monthly averages, yielding $n = 123$ observations—well above the minimum of 75 recommended for ARIMA-class models.

The goal of this project is to specify, fit, diagnose, and forecast a suitable time-series model for the monthly parallel rate. Along the way we annotate the series with the major political, economic, and public-health events that drove its most visible movements, providing an integrated quantitative-and-qualitative narrative of Algeria's parallel foreign-exchange market.

2 Model Specification

This section describes, step by step, the data-analysis pipeline used to arrive at our candidate models.

2.1 Data Loading and Preparation

The raw CSV file (`sellrate_timeseries.csv`) contains two columns: `date` (daily) and `sellRate`. Because the daily data exhibit a staircase pattern (many identical consecutive values, no intra-weekly seasonality, and $n > 3\,300$ observations), we aggregate to **monthly averages**. This yields a time-series object y_t of length $n = 123$ with `start = c(2016,1)` and `frequency = 12`.

2.2 Annotated Time Series with Key Events

The monthly series (Figure 1) shows:

- A pronounced **upward trend** spanning the entire period.
- Some visual evidence of mild **seasonality** (summer dips).
- Several **sharp** anomalies, most notably around 2020 (COVID-19 lockdown).

To contextualise the movements visible in the series, we overlay annotations for the main events known to have influenced Algeria's parallel exchange rate. annotations into four categories:

Financial-crisis shocks (red arrows)

Sep 2017 — PM Ouyahia authorised central-bank monetary financing ("money printing") amid a fiscal crisis (1). Fearing inflation, Algerians rushed to convert savings to euros, driving up the parallel EUR/DZD rate (1).

Jul 2025 — A record budget deficit (21.8 % of GDP), a USD 12 billion reserve drop, an import surge, and a rate cut pushed the parallel rate to historic highs (10). Prof. Mebtoul confirmed these structural pressures drove the unprecedented June–December 2025 black-market euro demand.

Policy and political shocks (purple arrows)

Apr–Jun 2019 — The Hirak movement sparked deep political uncertainty (2), and a wave of high-profile anti-corruption arrests followed, including five billionaires and, in June 2019, former Prime Minister Sellal (4). The parallel rate dipped visibly during this period. **No public source documents the reason for the dip**.

Oct 2022 — President Tebboune authorised used-car imports (Decree 23-74), triggering immediate speculative euro demand from anticipating importers (7).

Mar 2023 — Customs (DGD) operationalised the import scheme. Euro demand surged to fund the 2023 importation of 9 970 vehicles worth USD 142.4 million (3).

Oct 2023 — Car-import demand fully materialised, reaching USD 386.28 million total (including USD 243.88 million in H1 2024), sustaining elevated black-market euro rates (8).

Oct 2024 — Authorities abruptly suspended vehicle registrations under Decree 23-74 (9). Importers rushed to clear customs, creating a parallel euro supply panic that pushed rates sharply upward.

Dec 2024 and Jul 2025 — The government announced official travel allocations of EUR 750 per adult, effective 20 July 2025 (11). However, structural demand vastly exceeded this legal channel, failing to prevent new parallel-rate highs amid severe fiscal deterioration (10).

COVID-related upward moves (green arrows)

Jan 2020 — The WHO declared COVID-19 an international emergency, generating

initial precautionary euro demand (5).

Jun 2020 — The government began lifting containment measures (5). Restored economic activity expectations pushed euro demand upward ahead of anticipated cross-border commerce.

Oct 2020 — Eased domestic restrictions (5) and the global repeal of pandemic-era trade restrictions (6) signalled a return to normalcy, reviving latent euro demand.

Dec 2020 — Following the resumption of domestic flights, traders priced in international border reopenings, driving speculation that the euro could reach 250 DZD (5).

COVID-related downward moves (blue arrows)

Mar 2020 — A strict national lockdown following the WHO's pandemic declaration abruptly suppressed cross-border trade and travel, eliminating practical euro demand and depressing the rate (5).

Nov 2020 — Renewed restrictions and curfews disrupted parallel-market activity, again collapsing euro demand and depressing the rate (5).

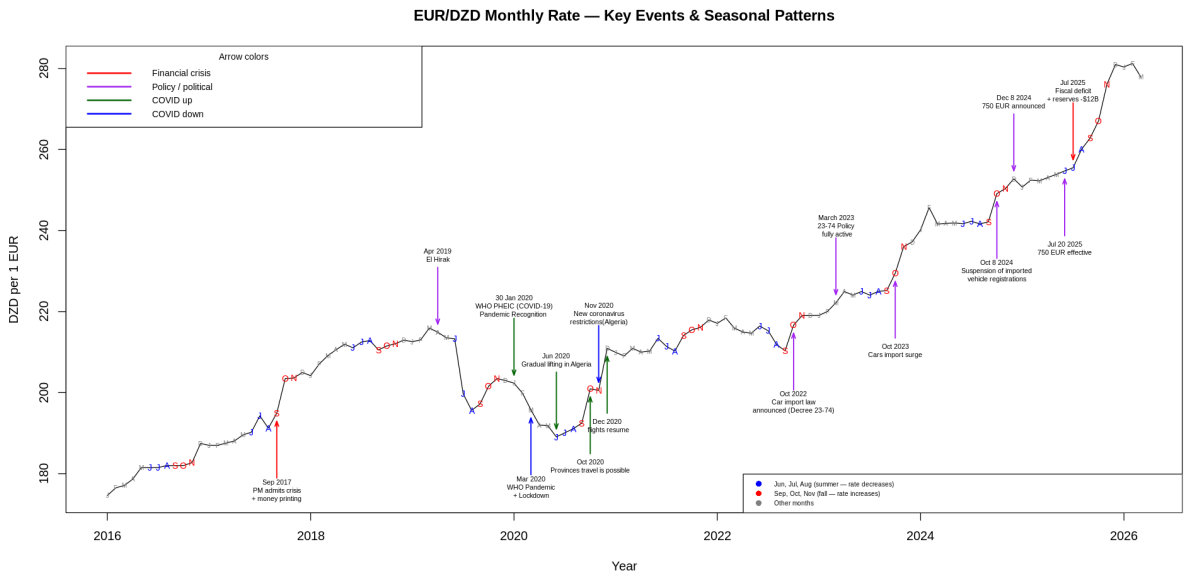

Figure 1: Monthly EUR/DZD rate annotated with key events.

2.3 Variance Stabilisation — Box–Cox Transformation

The Box–Cox family of power transformations is defined as

$$Y_t^{(\lambda)} = \frac{Y_t^{\lambda} - 1}{\lambda}, \qquad \lambda = 0 \implies \log Y_t.$$

We use `BoxCox.ar(y)` to estimate the optimal λ by maximising the profile log-likelihood. The resulting 95% confidence interval for λ ranges roughly from -0.5 to 1.7 and **contains**

$\lambda = 1$. Since $\lambda = 1$ lies inside the CI, no variance-stabilising transformation is needed; we proceed with the original series (Figure 2).

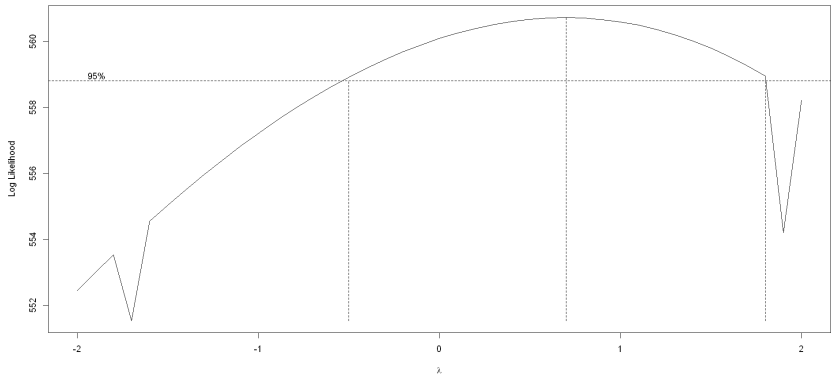

Figure 2: Box–Cox log-likelihood profile. The 95% CI (dashed) contains $\lambda = 1$.

2.4 Stationarity Testing — ADF Test

A time series Y_t is (weakly) stationary if $E[Y_t] = \mu$, $\mathrm{Var}(Y_t) = \sigma^2$, and $\mathrm{Cov}(Y_t, Y_{t+k})$ depends only on the lag k. The Augmented Dickey–Fuller (ADF) test evaluates:

$$H_0\colon \text{series has a unit root (non-stationary)} \qquad H_1\colon \text{series is stationary.}$$

Original series: ADF yields $p = 0.937 \gg 0.05$—we fail to reject H_0. The series is **not stationary**.

2.5 Differencing

We apply a first difference $\nabla Y_t = Y_t - Y_{t-1}$ to remove the linear trend. The differenced series (Figure 3) oscillates around zero, with a prominent negative spike in early 2020 (COVID shock).

Differenced series: ADF yields $p = 0.01 < 0.05$—we reject H_0. The differenced series **is stationary**. Therefore $d = 1$.

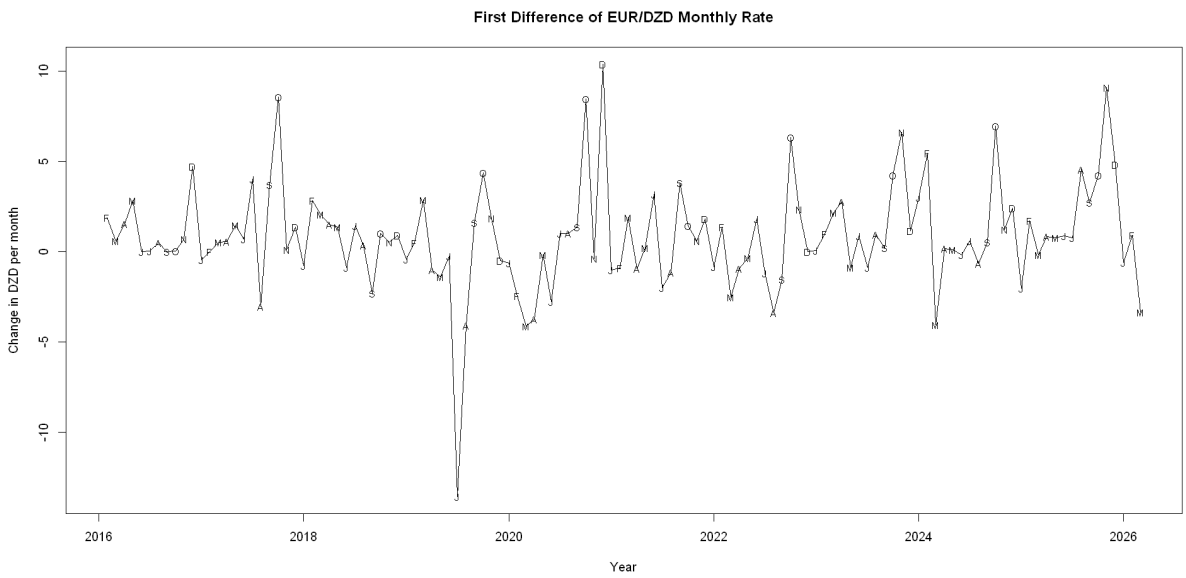

Figure 3: First difference ∇Y_t of the monthly EUR/DZD rate.

2.6 ACF and PACF Analysis

We inspect the sample ACF and PACF of the differenced series (Figure 4) to identify candidate ARMA orders:

The lag-1 spike in both functions suggests candidate models: **ARIMA(1,1,0)**, **ARIMA(0,1,1)**, or **ARIMA(1,1,1)**.

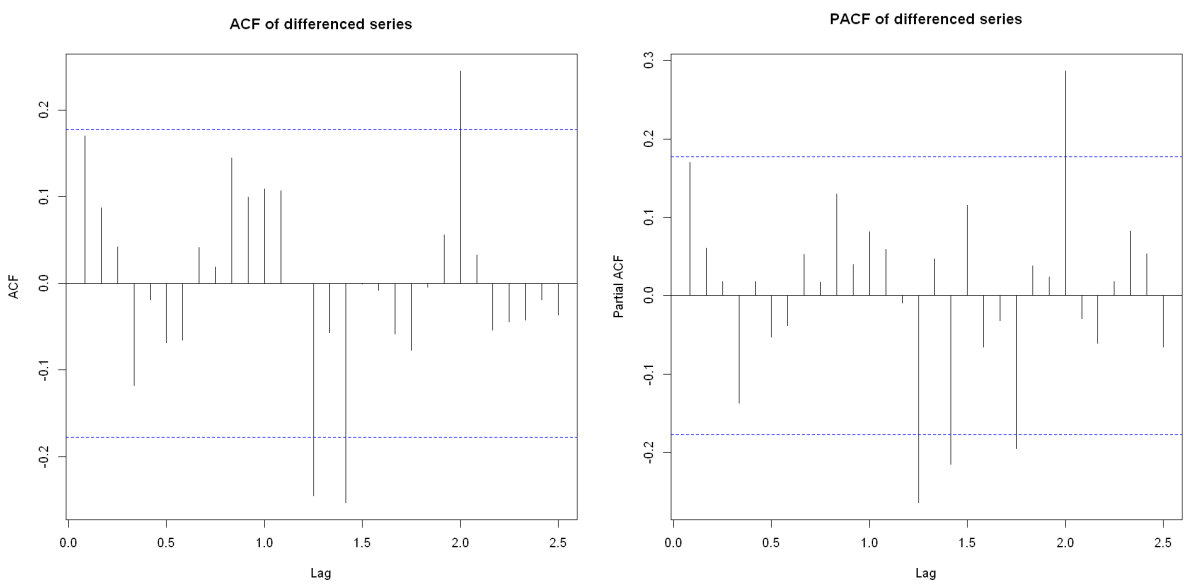

Figure 4: ACF (left) and PACF (right) of the differenced series.

- **ACF:** significant spike at lag 1; a few borderline spikes at seasonal lags 12 and 24; all other lags within the 95% confidence bands.

- **PACF:** significant spike at lag 1 that cuts off sharply; occasional borderline spikes at higher lags.

2.7 Automated Model Selection — `armasubsets`

The function `armasubsets(dy, nar=12, nma=12)` fits all possible ARMA(p, q) subsets and ranks them by the Bayesian Information Criterion:

$$\text{BIC} = -2 \log L + k \log n.$$

The best (darkest) row in Figure 5 selects only the intercept and Y-lag 1, corresponding to an **AR(1)** on the differenced series—i.e. ARIMA(1,1,0). No MA terms appear in the top model.

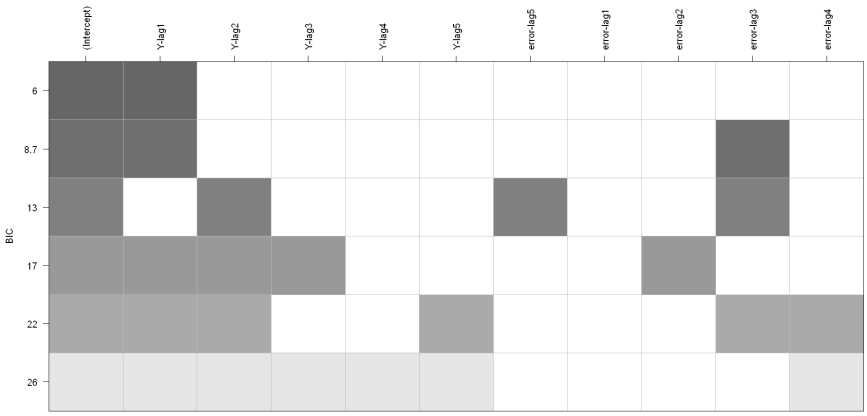

Figure 5: `armasubsets` BIC selection plot. Darkest row = lowest BIC.

2.8 Baseline Non-Seasonal Model

Based on the ACF/PACF behaviour and the BIC output from `armasubsets()`, the natural non-seasonal benchmark is **ARIMA(1,1,0)**. This model is statistically reasonable after first differencing, but the notebook diagnostics show that it does not fully capture the annual pattern in the data. In particular, the residual ACF/PACF still display structure at seasonal lags and the resulting forecasts quickly converge to a flat path. For that reason, ARIMA(1,1,0) is retained as a baseline rather than the final forecasting model.

3 Fitting and Diagnostics

3.1 Baseline ARIMA(1,1,0) Fit

The first fitted stochastic model is the non-seasonal benchmark **ARIMA(1,1,0)**. The estimated AR coefficient is $\phi_1 = 0.2365$, which is statistically significant and indicates

modest persistence in the differenced series. The model's estimated innovation variance is $\sigma^2 = 8.765$, with log-likelihood $\ell = -305.56$ and AIC $= 613.11$.

This benchmark captures the general upward movement of the series, but the fitted values lag behind the largest shocks and its residual ACF/PACF still show seasonal structure. The notebook's 10-step hold-out exercise also shows that the model quickly converges toward a flat forecast path, which is inappropriate for a series with visible annual seasonality.

3.2 Seasonal Extension to SARIMA(1,1,0)(1,1,0)$_{12}$

Because the non-seasonal benchmark leaves seasonal structure unexplained, the analysis extends to **SARIMA(1,1,0)(1,1,0)**$_{12}$. This specification adds one seasonal difference and a seasonal AR(1) term at lag 12:

$$\Phi(B^{12})\,(1-B)\,(1-B^{12})Y_t = \phi(B)\,\varepsilon_t, \qquad \varepsilon_t \sim \mathrm{WN}(0,\sigma^2).$$

The fitted coefficients are $\phi_1 = 0.1281$ and $\Phi_1 = -0.5910$, with $\sigma^2 = 9.807$, log-likelihood $\ell = -283.24$, and AIC $= 572.48$. Although the one-step innovation variance is slightly higher than in the ARIMA benchmark, the large AIC reduction and the markedly more realistic forecast shape justify retaining the seasonal model as the final forecasting specification.

3.3 Residual Diagnostics

Residual diagnostics are interpreted in two stages.

For **ARIMA(1,1,0)**, the residuals fluctuate around zero and the Ljung–Box test at lag 12 gives $p = 0.753$, so there is no evidence of short-run residual autocorrelation. However, the residual ACF/PACF still show spikes around seasonal lags, and the forecast evaluation confirms that the model misses the annual cycle.

For the final **SARIMA(1,1,0)(1,1,0)**$_{12}$ model, the residual time plot still contains large shocks, especially around the COVID period, but the remaining autocorrelation is no longer statistically significant at lag 12. The Q–Q plot and histogram show clear departures from normality, with heavy tails generated by the political, policy, and pandemic shocks highlighted earlier.

3.4 Formal Diagnostic Tests

Table 1: Key diagnostic results for the two fitted models.

Quantity	**ARIMA(1,1,0)**	**SARIMA(1,1,0)(1,1,0)$_{12}$**
AIC	613.11	572.48
Ljung–Box p-value (lag 12)	0.753	0.099
Shapiro–Wilk p-value	1.02×10^{-6}	2.19×10^{-7}

The main remaining deficiency is therefore **non-normality**, not residual autocorrelation. This is consistent with the empirical character of the series: the parallel exchange rate is repeatedly hit by rare but large shocks that no pure ARIMA-family model can anticipate.

Identified deficiencies of the final model:

- Residuals are not normally distributed because the series contains large exogenous shocks, especially around the pandemic period.
- The model does not account explicitly for intervention effects or regime changes such as the car-import policy shift and the 2025 fiscal stress.
- Prediction intervals should therefore be read as approximate, even though the seasonal dependence has been handled better than in the non-seasonal benchmark.

4 Forecasting

4.1 Forecast Results

The notebook compares the non-seasonal ARIMA benchmark with the seasonal SARIMA extension. The difference is qualitative as well as quantitative: ARIMA(1,1,0) produces a nearly flat forecast, whereas SARIMA(1,1,0)(1,1,0)$_{12}$ preserves the observed yearly rise-and-fall pattern.

Table 2: 10-step hold-out forecast accuracy comparison.

Model	**RMSE (DZD)**	**MAE (DZD)**
ARIMA(1,1,0)	18.74	15.67
SARIMA(1,1,0)(1,1,0)$_{12}$	9.31	8.13

The SARIMA model reduces the hold-out RMSE by about 50% relative to the ARIMA benchmark and lowers MAE from 15.67 to 8.13 DZD. This improvement is consistent

with the forecast plots: the seasonal model still underpredicts the late surge in the actual series, but it is far closer to the realised path than the flat non-seasonal forecast.

4.2 Forecast Plots

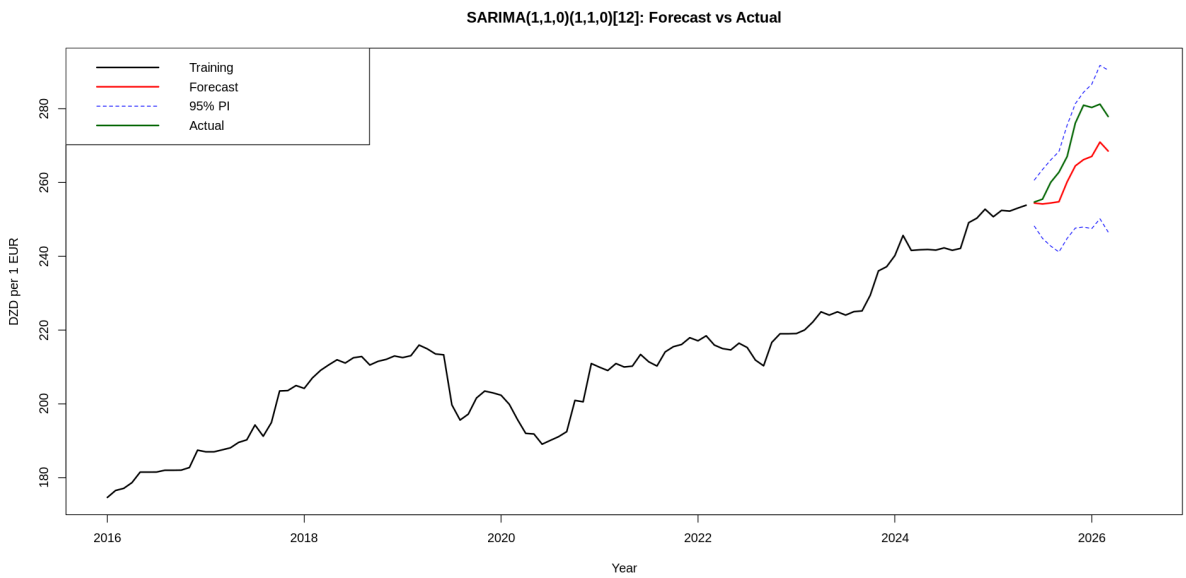

Figure 6: Full series with the 10-step SARIMA forecast

The SARIMA forecast is not perfect: it still underestimates the sharp late rise in the actual series, which is consistent with the notebook's broader conclusion that unexpected fiscal and policy shocks remain outside the reach of a purely autoregressive model. Nonetheless, it clearly dominates the non-seasonal ARIMA benchmark for practical forecasting.

5 Discussion

5.1 Main Conclusions

- The parallel EUR/DZD rate is better described by a **seasonal ARIMA structure** than by a purely non-seasonal ARIMA. ARIMA(1,1,0) remains a useful benchmark, but SARIMA(1,1,0)(1,1,0)$_{12}$ is the more credible forecasting model because it captures the yearly pattern instead of collapsing to a flat path.
- The series is driven as much by **exogenous shocks**—political upheavals, pandemic lockdowns, policy experiments (car imports, travel allowances), and fiscal imbalances—as by its own internal dynamics. A purely statistical model cannot anticipate these shocks, which is why even the SARIMA forecasts underperform during abrupt regime changes.

- The residuals remain **non-normal** because the market is subject to rare, large disturbances. This does not eliminate the value of the point forecasts, but it does mean prediction intervals should be interpreted cautiously.

5.2 Problems Encountered

- **Daily data quality:** the raw daily series contained many identical consecutive values (staircase pattern), making it unsuitable for direct modelling. Aggregation to monthly averages was the practical solution.
- **Structural breaks:** the car-import liberalisation (2022–2023), the 2024 registration freeze, and the 2025 deficit crisis represent regime changes that a standard ARIMA model cannot accommodate. Incorporating intervention variables or piecewise models could improve accuracy in future work.
- **Non-normality:** both ARIMA and SARIMA residuals fail the Shapiro–Wilk test because extreme shocks remain in the sample. While this does not invalidate the model's point forecasts, it makes Gaussian prediction intervals only approximate.
- **Forecast sensitivity:** the seasonal model improves average accuracy, but any sudden policy change or political event can still create large forecast errors over short horizons.

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